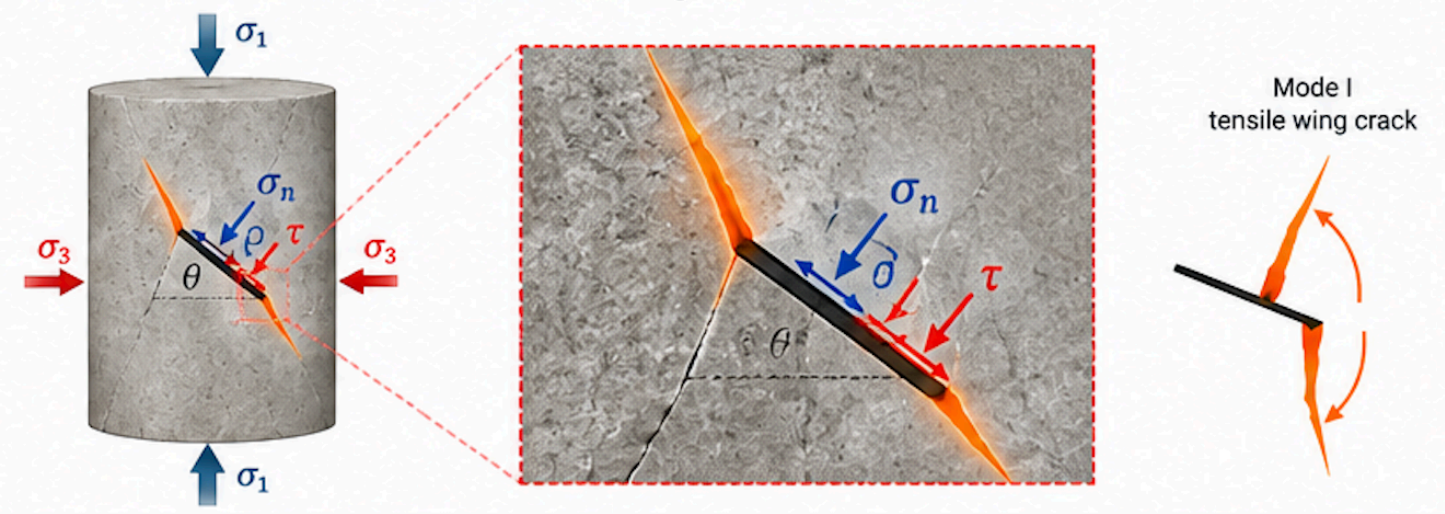
Why do tensile microcracks form in compression?

An inclined flaw concentrates local stress.



Mechanistic interpretation

Sliding along an existing inclined defect produces tensile stress concentrations near its tips. Wing cracks can extend approximately toward the loading direction.

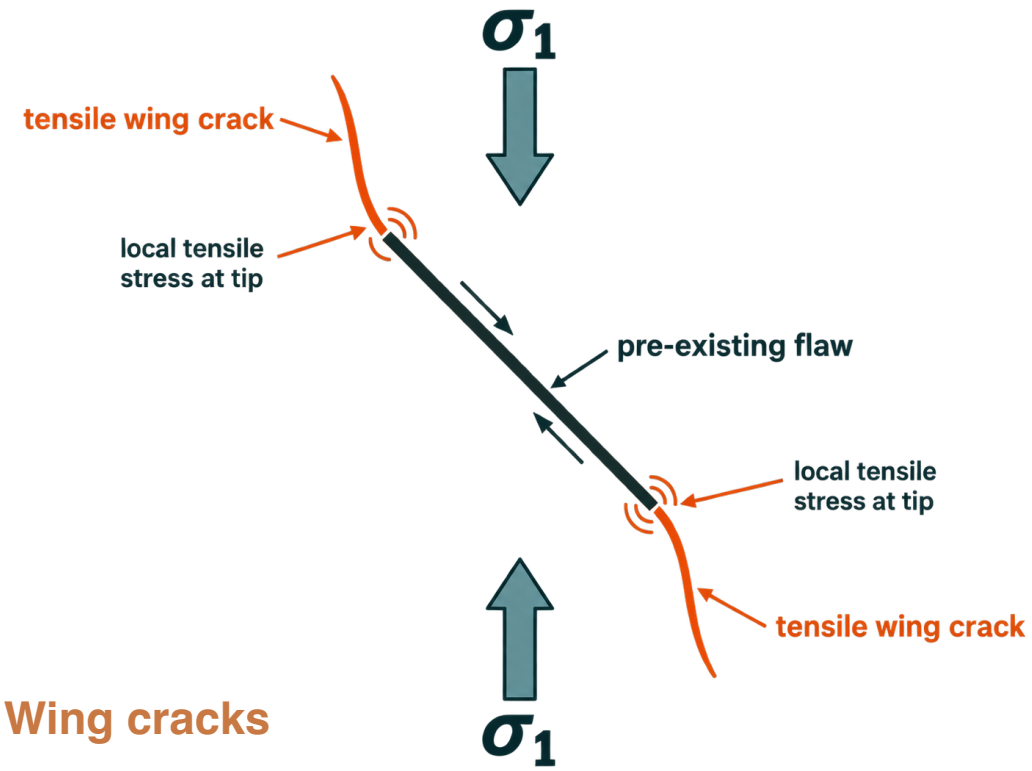
This is a model—not every observed fracture is a simple wing crack.

Wing cracks

Compressive loading ≠ exclusively shear-mode cracking.

Why do tensile microcracks form in compression?

An inclined flaw concentrates local stress.



Wing cracks

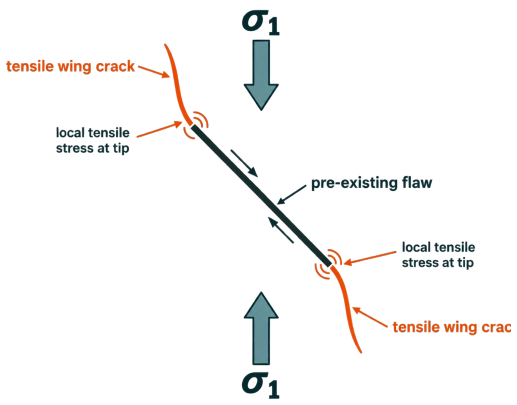
Physical sequence

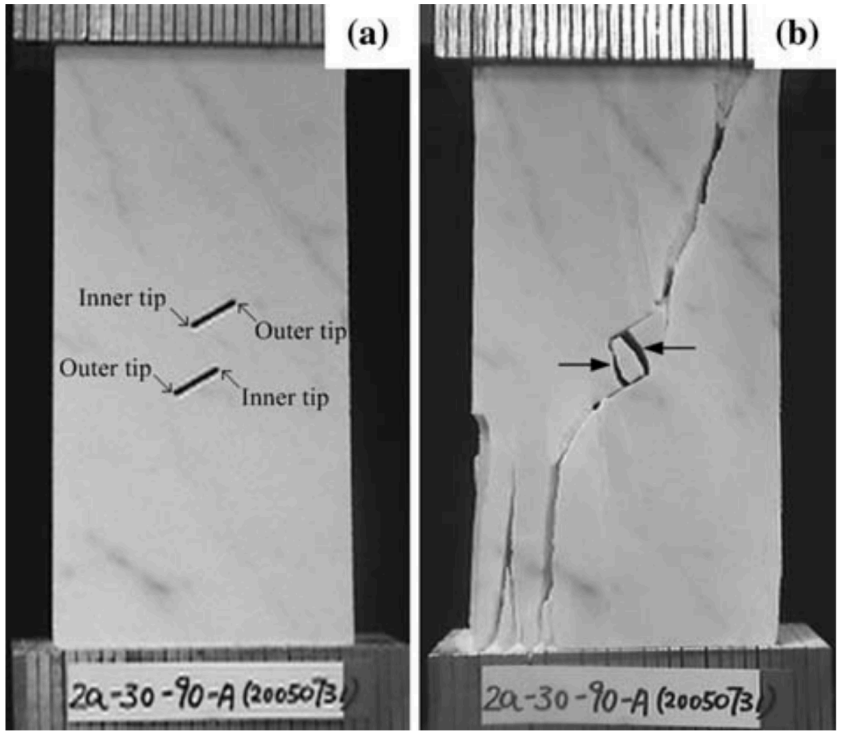
1. Pre-existing flaw is inclined to σ_1.
2. Resolved shear stress activates sliding.
3. Tensile wing cracks grow from the tips, tending to align with σ_1.

Compressive loading ≠ exclusively shear-mode cracking.

Why do tensile microcracks form in compression?

An inclined flaw concentrates local stress.





Wing cracks

Physical sequence

1. Pre-existing flaw is inclined to σ_1.
2. Resolved shear stress activates sliding.
3. Tensile wing cracks grow from the tips, tending to align with σ_1.

Compressive loading ≠ exclusively shear-mode cracking.

Wong, L. N. Y., & Einstein, H. H. (2009). Crack Coalescence in Molded Gypsum and Carrara Marble: Part 1. Macroscopic Observations and Interpretation. *Rock Mechanics and Rock Engineering*, *42*(3), 475–511. https://doi.org/10.1007/s00603-008-0002-4

Sliding begins only when shear overcomes friction

$$|\tau| > \mu\,\sigma_n$$

Higher differential stress

Raises the resolved shear traction and activates more flaws.

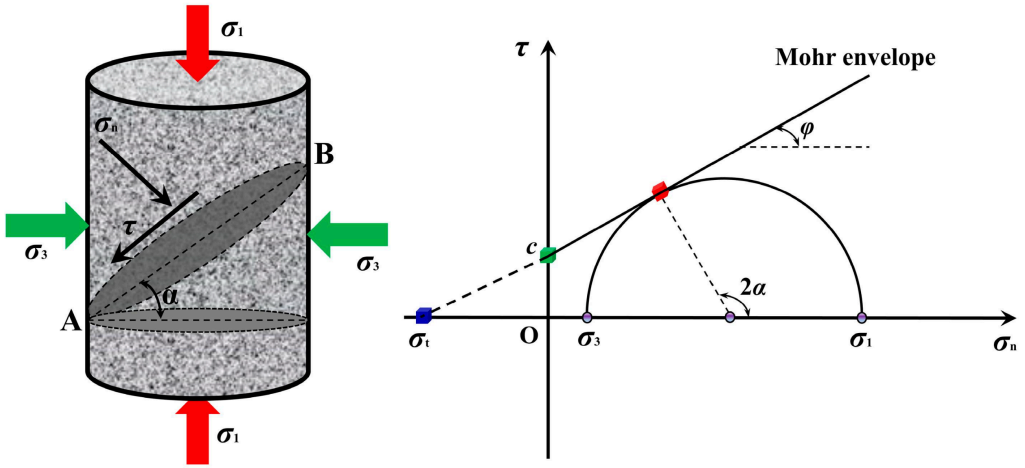
Higher friction μ

Delays slip and raises the stress needed for wing-crack nucleation.

Higher confinement σ_3

Raises normal clamping and suppresses crack opening.

Slip is the source event. Wing-crack opening is the consequence at the tips.



A wing crack starts when local mode-I loading stress intensity factor (K_i) reaches the fracture toughness of the rock (K_{ic})

$$K_i \geq K_{ic}$$

K_i

Tip-scale driving force produced by sliding, geometry and the applied stresses. **Intensity** of the opening-mode **stress field near the crack tip**.

K_{ic}

Material resistance to opening-crack growth: mode-I fracture toughness.

How strongly are we driving the crack to open and propagate?

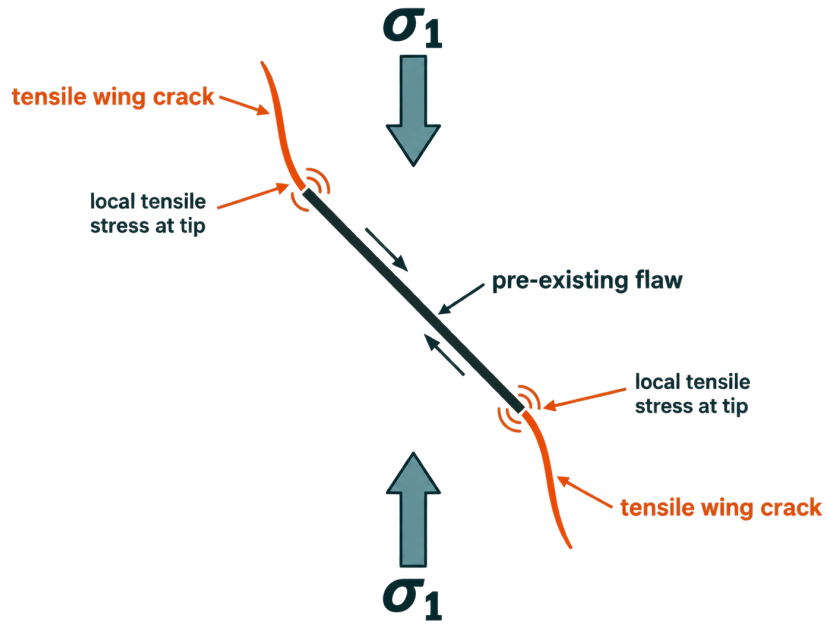
crack of half-length (a) in an elastic material

$$K_I = Y\sigma\sqrt{\pi a}$$

(σ) represents the remote tensile stress opening the crack

(Y) is a dimensionless factor accounting for geometry

How much crack-tip loading can the material resist before the crack extends?



This lecture uses the criterion qualitatively. Full stress-intensity derivations are outside scope.

How each control changes crack activation

CONTROL	DIRECT PHYSICAL EFFECT	EXPECTED RESPONSE
More confinement	Normal traction on the parent flaw increases. Compression across the wing also increases.	Sliding and wing nucleation require higher σ_1. At the same σ_1, wings remain shorter.
Higher friction	The resistance $\mu\sigma_n$ increases for every closed flaw.	First sliding and wing nucleation occur later. Peak strength increases.
Longer initial flaw	For comparable geometry, K_i scales with applied stress times $\sqrt{c}$.	The nucleation stress decreases. The interaction length scale becomes larger.
Flaw parallel to σ_1	The resolved shear traction vanishes because $\sin 2\beta = 0$.	No sliding occurs in the idealized model, so this flaw does not nucleate a wing crack.

These predictions compare one change at a time while geometry and material toughness remain fixed.

Confinement strengthens rock by clamping and crack closure

On the parent flaw

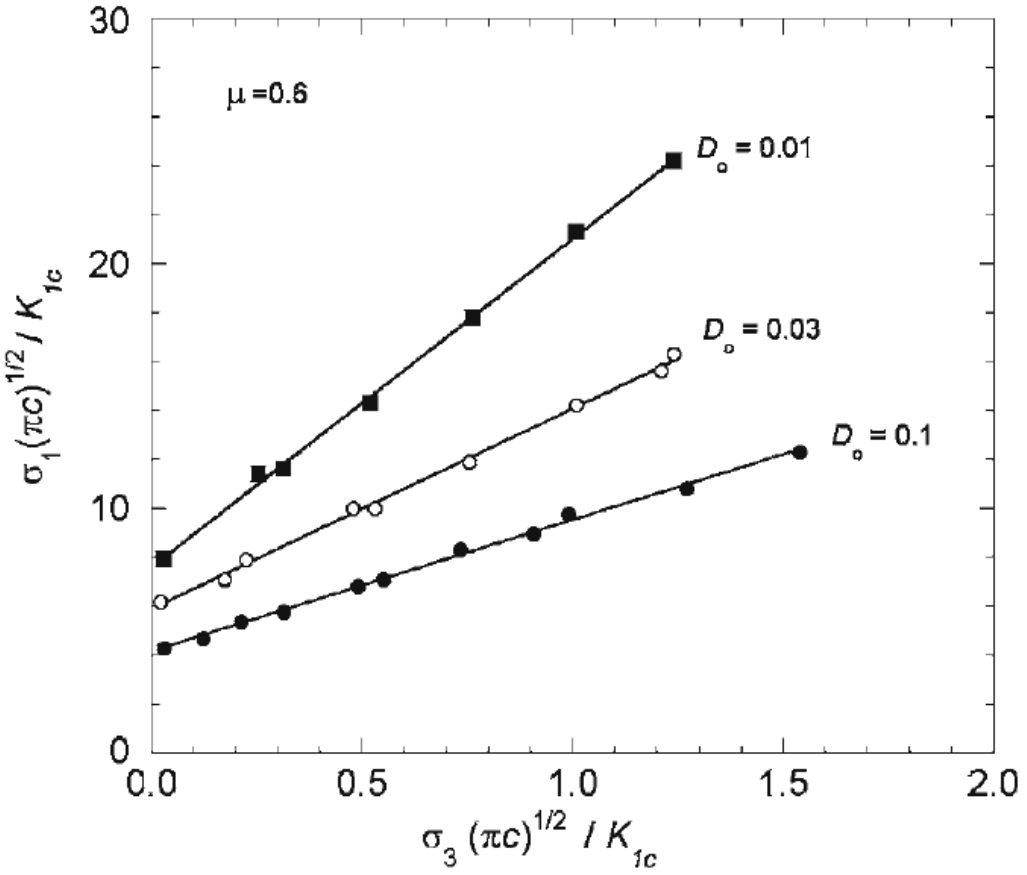
σ_3 raises normal traction. Frictional resistance $\mu\sigma_n$ increases.

On the wing crack

Compression across the wing opposes opening and promotes arrest.

How to read the model graph

Horizontal axis: normalized confinement σ_3
Vertical axis: normalized peak stress σ_1
D_0: initial crack damage, proportional to crack density and crack size²

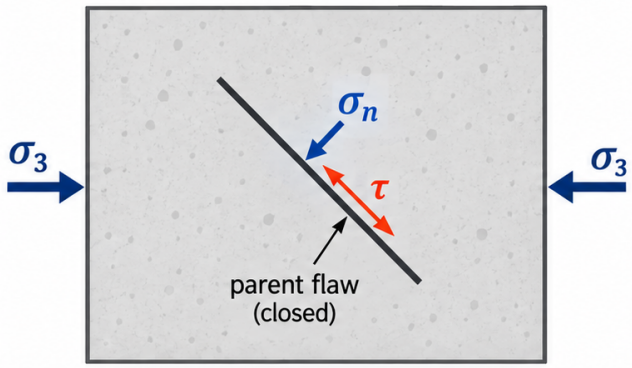


The same crack population therefore needs a larger σ_1 at larger σ_3.

Paterson & Wong (2005), Fig. 50; Ashby & Sammis (1990)

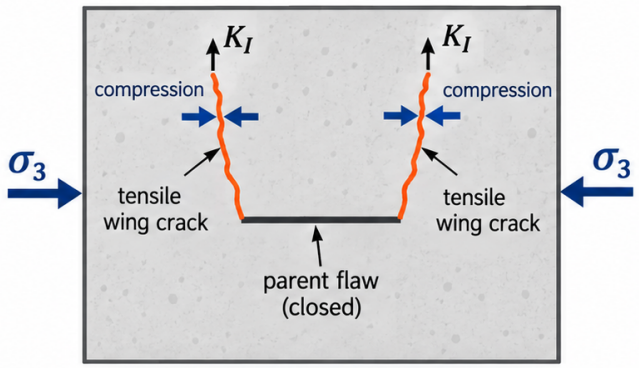
Why confinement raises peak strength

1 Parent flaw



Higher σ_3 raises σ_n and therefore increases the frictional resistance $\mu\sigma_n$. Sliding starts at a larger σ_1.

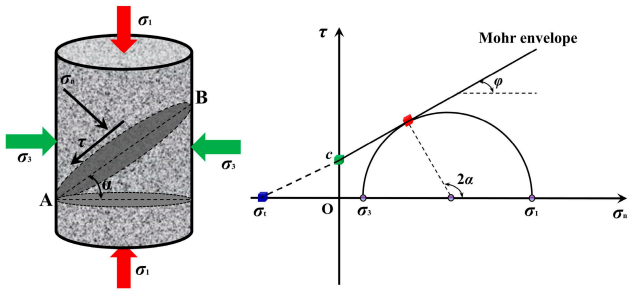
2 Wing crack

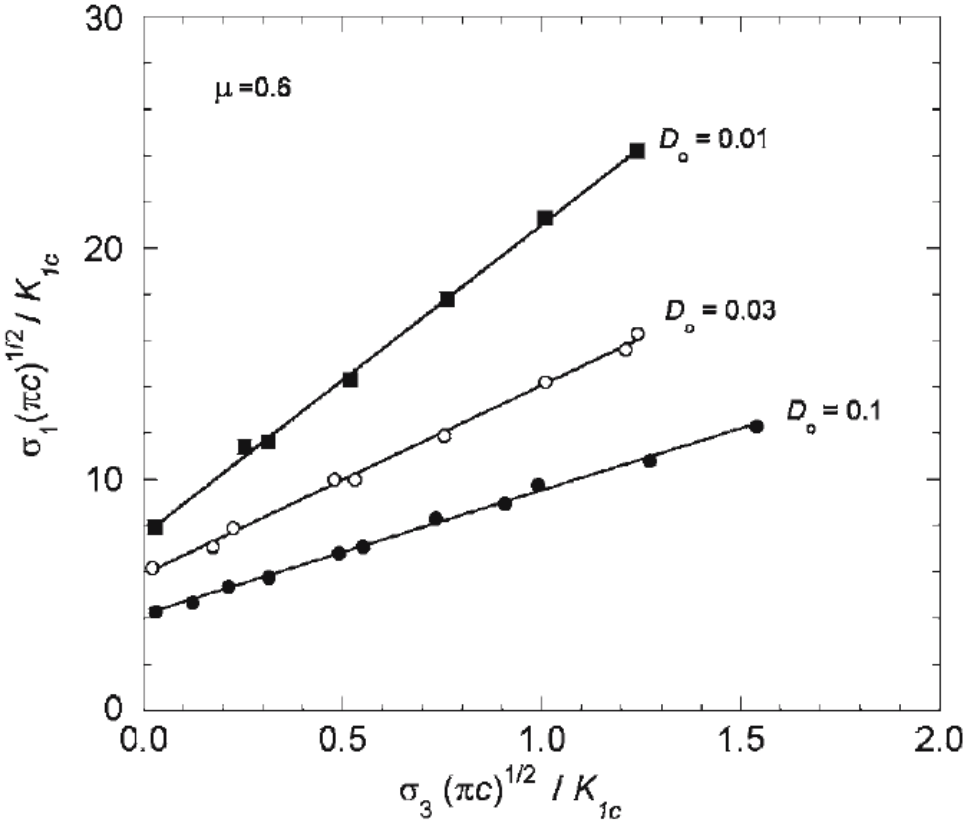


Higher σ_3 compresses the wing-crack plane; reduces the opening drive K_I and promotes stable arrest.

$$\sigma_n = (\sigma_1 + \sigma_3)/2 - (\sigma_1 - \sigma_3)\cos 2\beta / 2$$

$$\tau = (\sigma_1 - \sigma_3)\sin 2\beta / 2$$

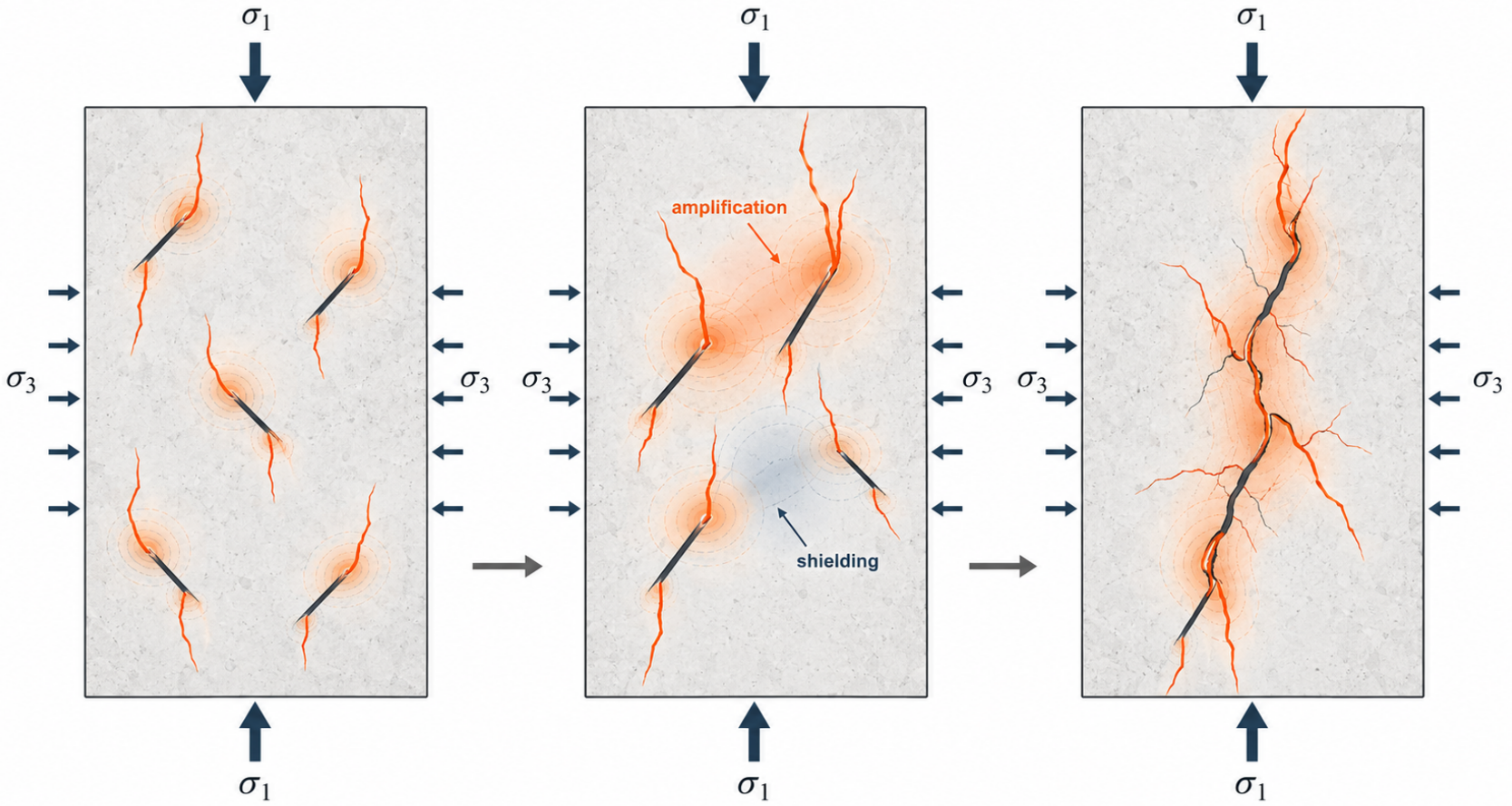




Along every D_0 line, more confinement requires a larger σ_1 for instability. Lower initial damage gives the stronger envelope.

Paterson & Wong (2005), Fig. 50; model after Ashby & Sammis (1990)

Macroscopic failure requires a population of interacting cracks



Stress amplification

One crack changes the local driving force on its neighbours.

Shielding

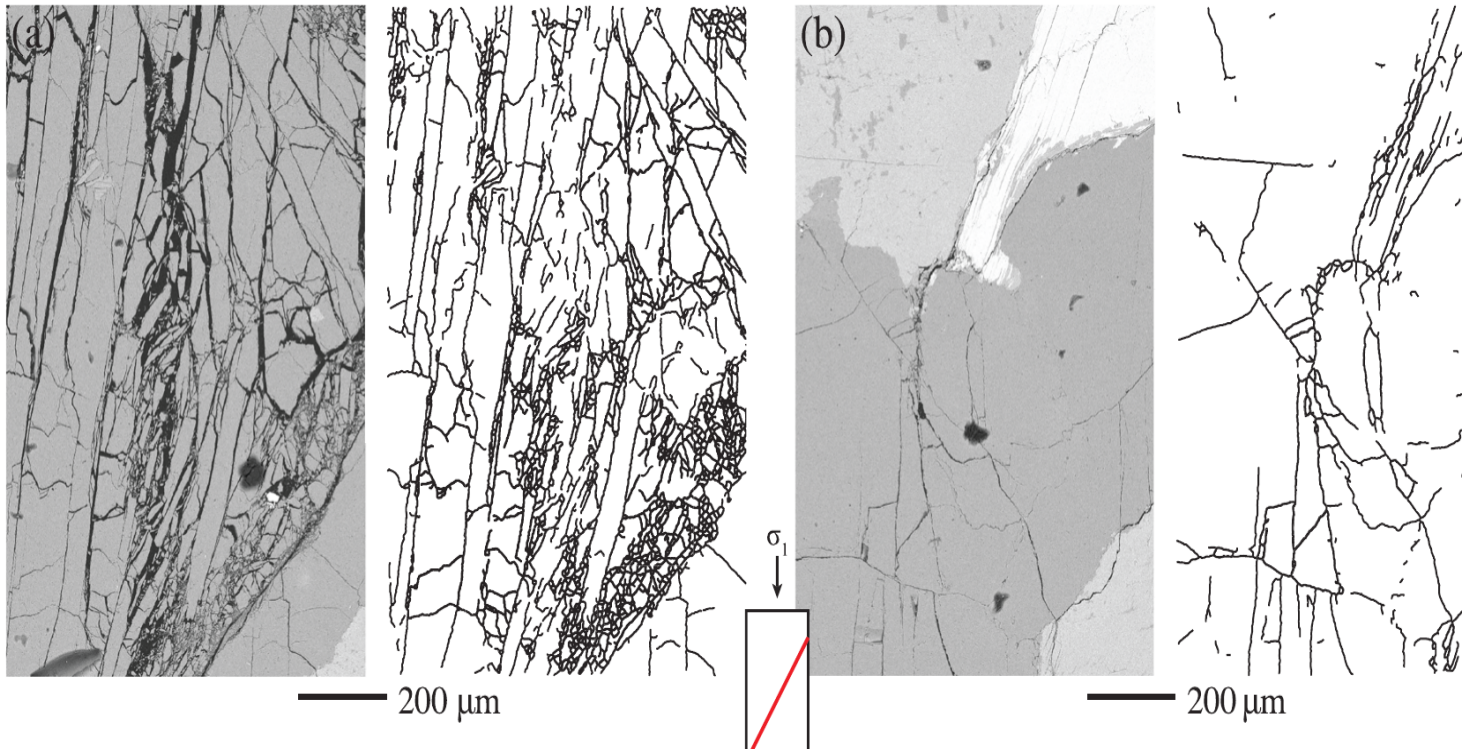
Some arrangements reduce local intensity and delay linkage.

Coalescence

Favourable arrangements bridge ligaments and enlarge the damage scale.

Paterson & Wong (2005), Fig. 48

Individual cracks interact and coalesce



Three stages

1 Isolated crack growth

2 Crack-tip interactions and linkage

3 Connected fracture path and localization

Coalescence—not one perfectly continuous growing crack—commonly creates the fault.